Project Initialization and Planning Phase

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

Develop a supervised classification model to Minimize financial loss by reducing:

False Negatives (missed fraud)

Fraud exposure window

Customer trust erosion

Project Overview	
Objective	Develop a supervised classification model to detect fraudulent transactions while minimizing False Negatives.
Scope	The scope of this project includes: Designing a supervised classification model using structured transactional data. Performing data preprocessing, feature engineering, and imbalance analysis. Benchmarking multiple tree-based machine learning models. Selecting a model based on recall-focused evaluation metrics. Serializing the trained model for deployment. Building a prototype web interface for real-time fraud prediction.
Problem Statement	

Description	Online financial transactions are increasingly vulnerable to fraudulent activities such as unauthorized transfers, rapid cash-outs, and account manipulation. Traditional rule-based fraud detection systems struggle to adapt to evolving fraud patterns and often generate high false positive or false negative rates. The challenge lies in developing a robust classification model capable of identifying fraudulent transactions within a highly imbalanced dataset (fraud cases ≈ 0.13%) while minimizing missed fraud instances. This problem is fundamentally a cost-sensitive classification task, where the cost of misclassifying fraudulent transactions is significantly higher than incorrectly flagging legitimate ones.
Impact	Solving this problem provides measurable value across multiple dimensions: Financial Impact Reduces direct monetary losses from undetected fraud. Minimizes compensation and chargeback costs. Operational Impact Improves fraud response efficiency. Reduces dependency on manual rule-based systems. Customer Impact Enhances trust in digital payment systems. Prevents account compromise and unauthorized transactions. Strategic Impact Enables scalable, adaptive fraud detection systems. Supports data-driven risk management decisions.
Proposed Solution	

Approach	The solution follows a structured machine learning lifecycle: Data Understanding & Cleaning Remove irrelevant identifiers. Analyze class imbalance and transaction patterns. Feature Engineering Encode categorical transaction types. Normalize numerical features where required. Retain balance-related variables critical to fraud detection. Model Development Train multiple tree-based classifiers: Decision Tree Random Forest Extra Trees Compare models using recall-focused evaluation. Model Selection Prioritize fraud recall and confusion matrix analysis. Select model with lowest False Negatives. Model Serialization Save final model using pickle for deployment. Deployment Prototype Develop a Streamlit-based web application. Enable real-time transaction prediction via user input.

Key Features	Recall-Optimized Design Model selection prioritizes fraud detection sensitivity over raw accuracy. Imbalance-Aware Evaluation Recognizes that accuracy is misleading for highly skewed datasets. End-to-End ML Pipeline Covers full lifecycle: data preprocessing → training → evaluation → deployment. Interpretability Decision Tree model provides explainable classification logic. Scalable Architecture (Prototype Level) Easily extendable to REST API–based production systems. Business-Aligned Metrics Focuses on minimizing financial exposure rather than maximizing superficial metrics.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Processing capability for model training and experimentation	Google Colab CPU runtime (2–4 virtual cores)
Memory	RAM Required to handle ~6.3M transactional records during preprocessing	12-16 GB RAM
Storage	Required for dataset, model artifacts, and logs	~500 MB (dataset ~178 MB, model < 50 MB, logs minimal)
Software		
Frameworks	Python Web deployment frameworks	Streamlit

Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, pickle, pyngrok
Development Environment	IDE	Google Colab
Data		
Data	Source, size, format	Kaggle – Online Payments Fraud Detection Dataset; 6,362,620 rows; CSV